

layout: post title: SwitchFree library prep date: '2024-04-05' categories: tags: [DNA, RNA, Library prep, Protocols]

projects: Kit test

SwitchFree library prep test

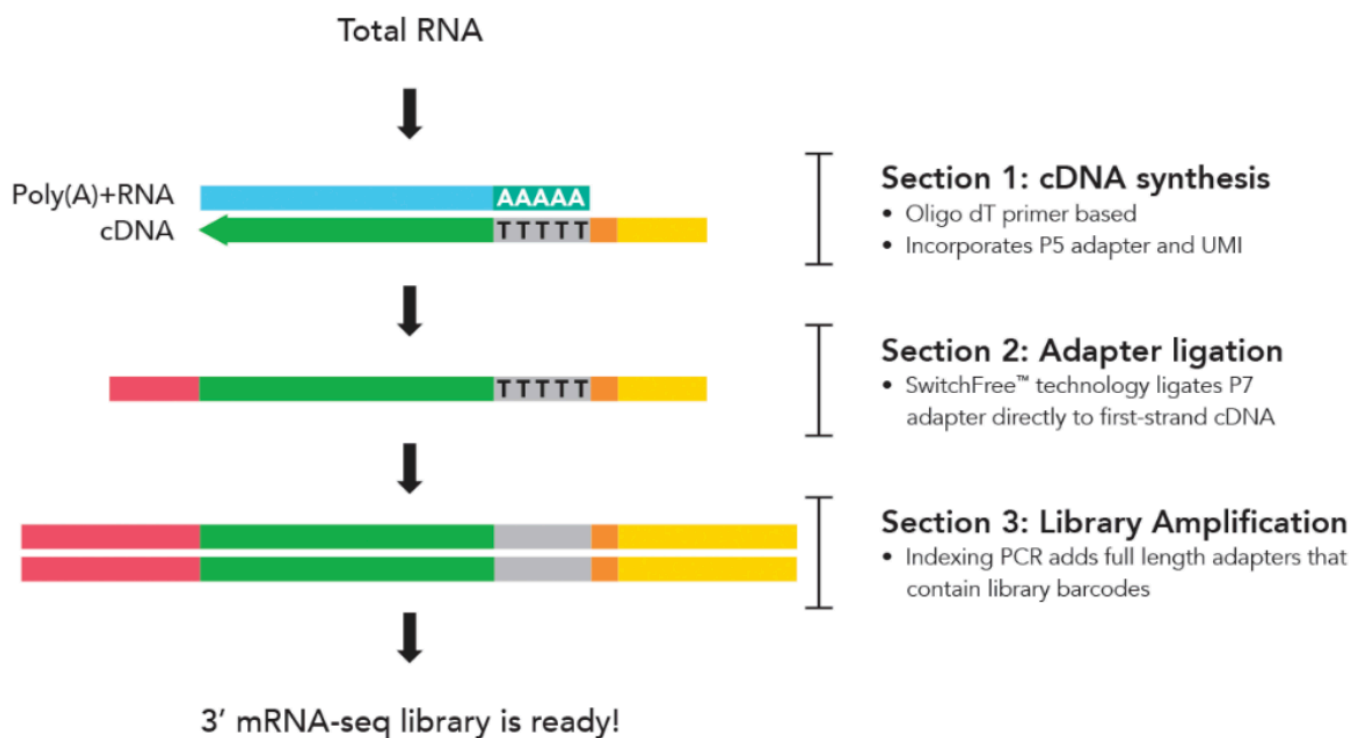
This post details the info about the SwitchFree library prep steps. Hollie got this kit as a test to see if we can produce adequate libraries for RNA sequencing with different species. I'm using the [Zymo-Seq SwitchFree 3' mRNA Library Kit](#). See Zymo's protocol [here](#).

We will be using samples 8, 9, 106, 109, 111, and 112 from the POC 2023 spawning project. These samples were extracted on [4/3/24](#). The kit needs a minimum of 10 ng of total RNA or a maximum of 500 ng of total RNA, which is a large range. Since some of the concentrations in these samples were low, we will be using 11 ng of total RNA as input. Here's a breakdown of input RNA volumes for each sample:

TubeID	TS RNA (ng/uL)	Strip tube #	RNA (uL)	Ultrapure water (uL)	Total starting volume (ul)	Primer
8	15.9	1	0.7	4.3	5.0	7
9	23.2	2	0.5	4.5	5.0	8
106	7.29	3	1.5	3.5	5.0	9
109	8.71	4	1.3	3.7	5.0	10
111	7.23	5	1.5	3.5	5.0	11
112	6.27	6	1.8	3.2	5.0	12

Here's the SwitchFree library prep workflow:

Zymo-Seq SwitchFree™ 3' mRNA Library Kit Workflow



Materials

- [Zymo-Seq SwitchFree 3' mRNA Library Kit](#)
- PCR tubes
- Thermocycler
- Mini centrifuge
- Aluminum beads (to keep things on ice)
- Magnetic stand for PCR tubes
- Kit contents

Zymo-Seq SwitchFree™ 3' mRNA Library Kit	R3008 (12 preps)	R3009 (96 preps)	Storage Temperature
PolyA R1 Reagent	60 µL	60 µL × 8	-80 °C
R2 Reagent	120 µL	120 µL × 8	-80 °C
Reaction Clean-up Solution	50 µL	50 µL × 8	-80 °C
3'mRNA L Reagent	180 µL	180 µL × 8	-80 °C
ZymoTaq™ PreMix	300 µL	300 µL × 8	-80 °C
Zymo-Seq™ UDI Primer Sets (Indexes 1-12) ¹	20 µL / Index	-	-80 °C
Zymo-Seq™ UDI Primer Plate (Indexes 1-96) ²	-	10 µL / Index	-80 °C
Select-a-Size MagBead Concentrate	300 µL	1.5 mL	4 °C
Select-a-Size MagBead Buffer	10 mL	50 mL	4 °C
Library Binding Solution	1 mL	10 mL	4 °C
DNA Wash Buffer (concentrate)	6 mL	48 mL	Room Temp.
DNA Elution Buffer	1 mL x 2	16 mL	Room Temp.
DNase/RNase-Free Water	1 mL × 2	10 mL	Room Temp.
Instruction Manual	1	1	-

Buffer preparation

Once buffers are prepared for a kit, they do not need to be prepared again.

- Add 300 µL of the Select-a-Size MagBead concentrate to 10 mL of Select-a-Size MagBead Buffer.
 - These should be prepared at least 5 days ahead of library prep.
- Add 24 mL of 100% ethanol to the DNA Wash Buffer concentrate. Store at room temperature.

Best practices

- Avoid multiple freeze thaws of all components, and aliquot components as necessary
- Remove enzymes from cold storage just before use and return to cold storage immediately after use
- Thaw and maintain all components on ice unless noted otherwise
- Flick to mix thawed components and briefly centrifuge prior to use
- After adding each component, mix by pipetting up and down 15-20 times. Briefly centrifuge after
- Pre-program thermal cycler with lid heating ON set to >100-105°C
- Turn on thermocycler day-of to preheat
- Pre-calculate how much to dilute RNA samples

Protocol

Protocol was followed according to this [post](#).

Sections 1 and 2 were done on 4/4/24. Section 3 was done on 4/5/24.

QC

Run [DNA Tapestation](#) to visualize libraries. Here's an example of what the library should look like on a Tapestation:

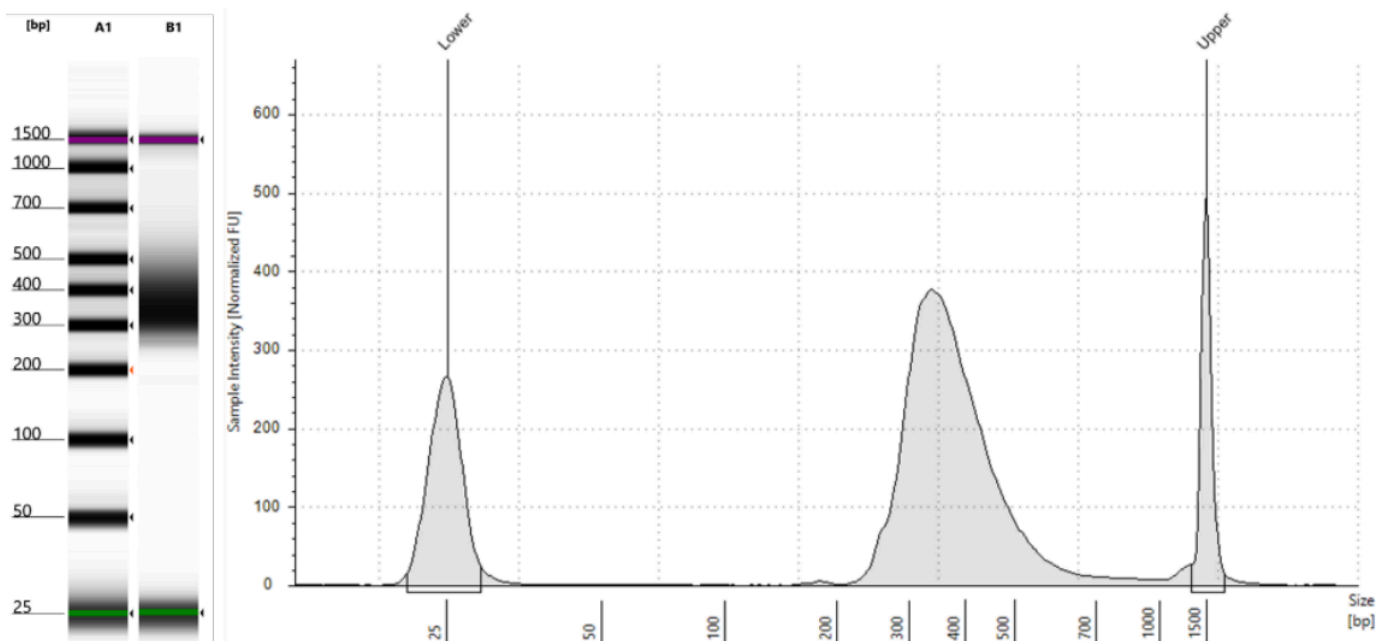
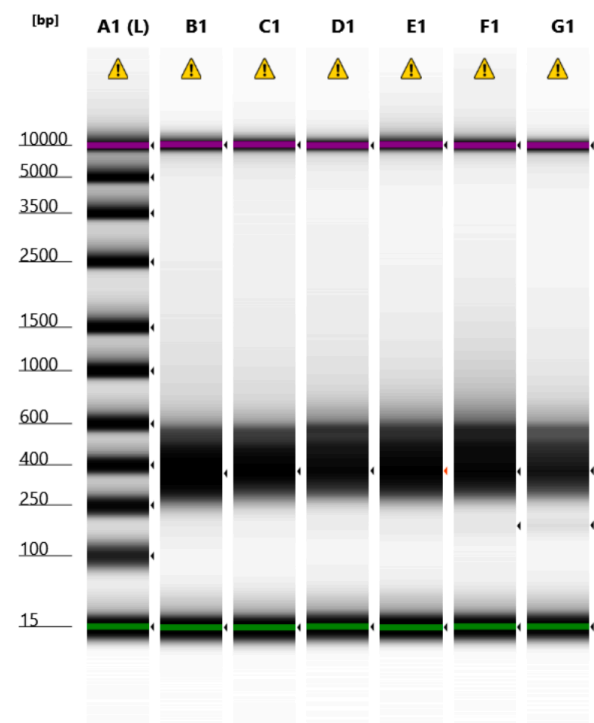


Figure 1. Agilent 4150 TapeStation® HSD1000 electropherogram of a typical Zymo-Seq SwitchFree™ 3' mRNA Library using 10 ng of HeLa total RNA (RIN > 9) and indexed using 19 PCR cycles. Fragment sizes mainly range from 250 to 700 bp. The peaks at 25 and 1500 bp are the interior standards of the High Sensitivity D1000 tape.

I ran the tapestation on the libraries on 4/5/24. See the full report [here](#).

Filename: 2024-04-05 - 11.46.53.D5000

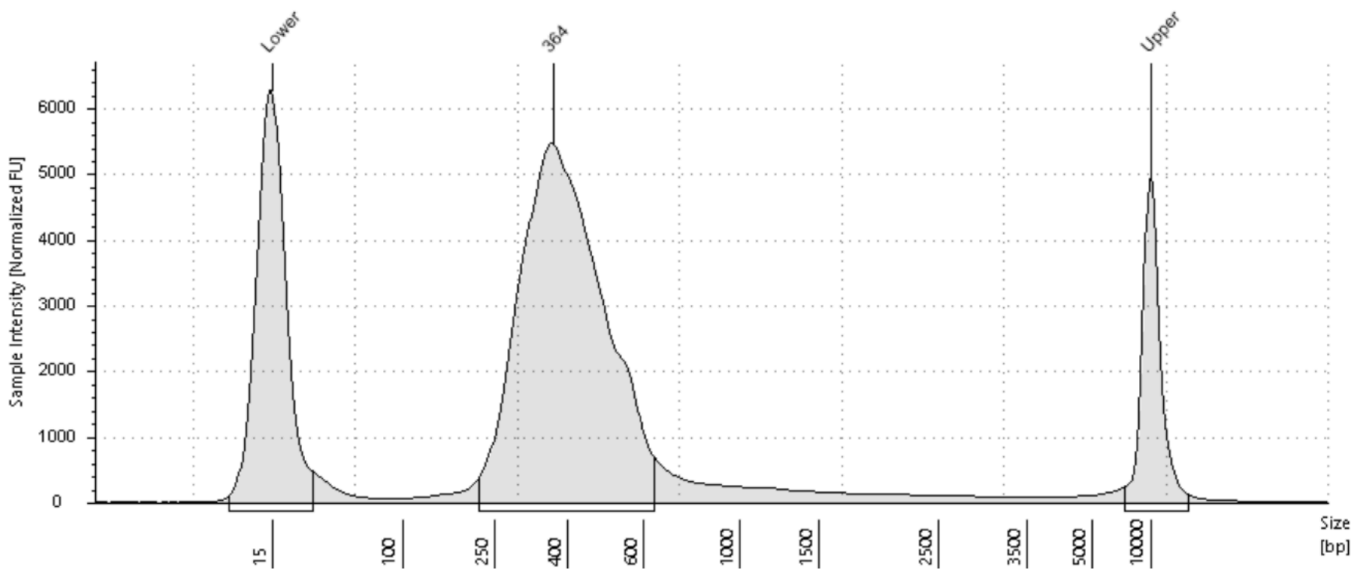


Default image (Contrast 100%)

Sample Info

Well	Conc. [ng/μl]	Sample Description	Alert	Observations
A1	46.2	Ladder	⚠	Caution! Expired ScreenTape device; Ladder
B1	19.3	8	⚠	Caution! Expired ScreenTape device
C1	16.6	9	⚠	Caution! Expired ScreenTape device
D1	14.2	106	⚠	Caution! Expired ScreenTape device
E1	18.1	109	⚠	Caution! Expired ScreenTape device
F1	15.5	111	⚠	Caution! Expired ScreenTape device
G1	12.3	112	⚠	Caution! Expired ScreenTape device

B1: 8



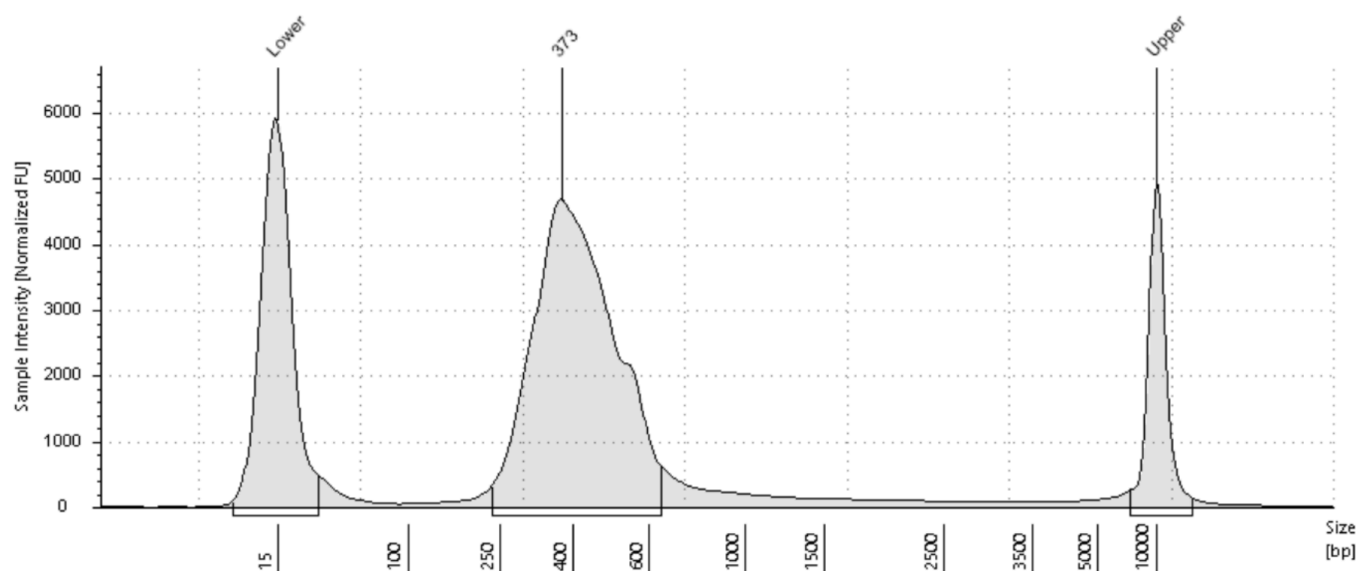
Sample Table

Well	Conc. [ng/μl]	Sample Description	Alert	Observations
B1	19.3	8		Caution! Expired ScreenTape device

Peak Table

Size [bp]	Calibrated Conc. [ng/μl]	Assigned Conc. [ng/μl]	Peak Molarity [nmol/l]	%Integrated Area	Peak Comment	Observations
15	6.87	-	705	-		Lower Marker
364	19.3	-	81.6	100.00		
10000	3.25	3.25	0.500	-		Upper Marker

C1: 9



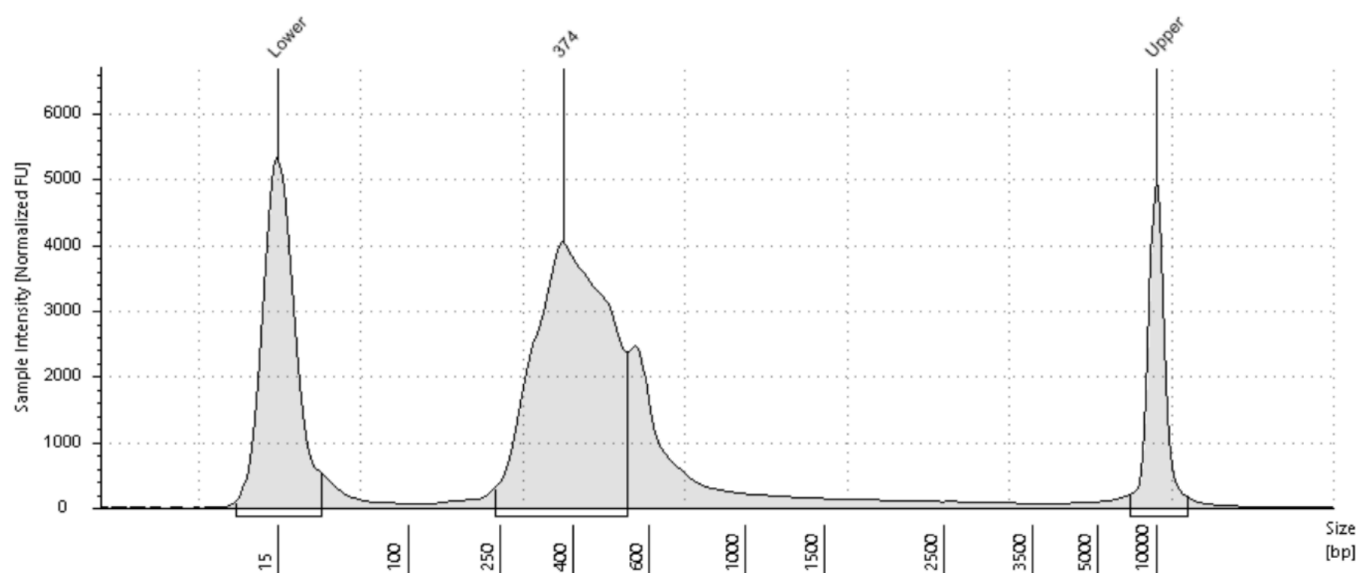
Sample Table

Well	Conc. [ng/μl]	Sample Description	Alert	Observations
C1	16.6	9	⚠	Caution! Expired ScreenTape device

Peak Table

Size [bp]	Calibrated Conc. [ng/μl]	Assigned Conc. [ng/μl]	Peak Molarity [nmol/l]	% Integrated Area	Peak Comment	Observations
15	6.97	-	715	-		Lower Marker
373	16.6	-	68.5	100.00		
10000	3.25	3.25	0.500	-		Upper Marker

D1: 106



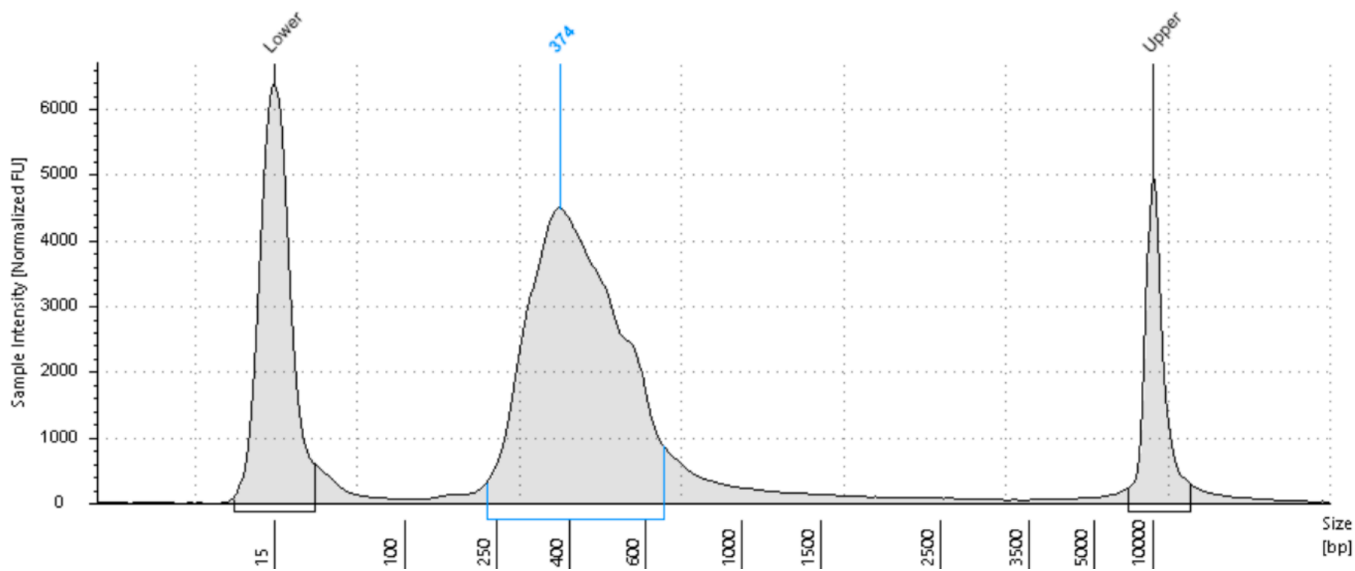
Sample Table

Well	Conc. [ng/μl]	Sample Description	Alert	Observations
D1	14.2	106	⚠	Caution! Expired ScreenTape device

Peak Table

Size [bp]	Calibrated Conc. [ng/μl]	Assigned Conc. [ng/μl]	Peak Molarity [nmol/l]	% Integrated Area	Peak Comment	Observations
15	6.85	-	703	-		Lower Marker
374	14.2	-	58.6	100.00		
10000	3.25	3.25	0.500	-		Upper Marker

E1: 109



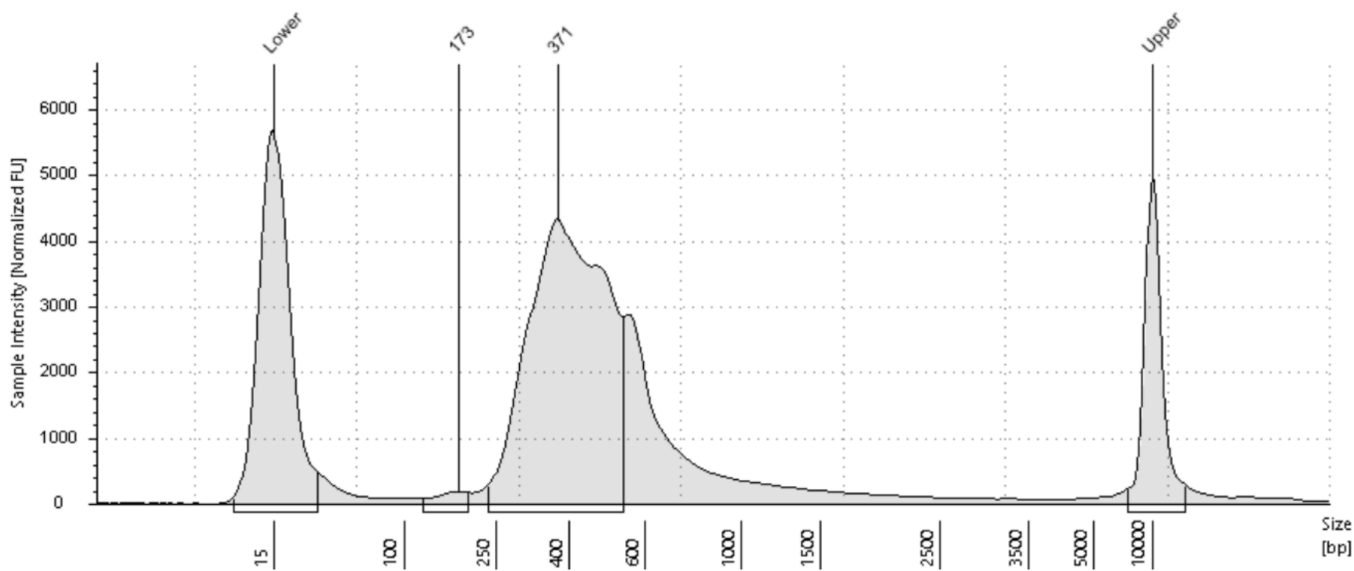
Sample Table

Well	Conc. [ng/μl]	Sample Description	Alert	Observations
E1	18.1	109		Caution! Expired ScreenTape device

Peak Table

Size [bp]	Calibrated Conc. [ng/μl]	Assigned Conc. [ng/μl]	Peak Molarity [nmol/l]	% Integrated Area	Peak Comment	Observations
15	7.39	-	758	-		Lower Marker
374	18.1	-	74.4	100.00		
10000	3.25	3.25	0.500	-		Upper Marker

F1: 111



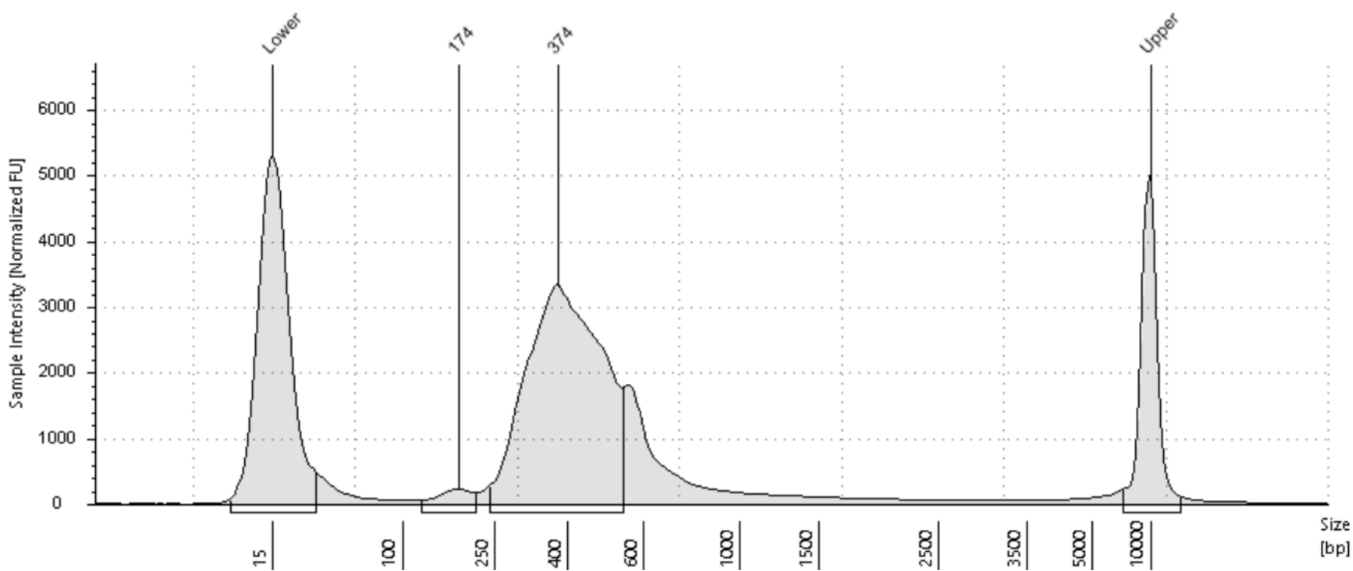
Sample Table

Well	Conc. [ng/μl]	Sample Description	Alert	Observations
F1	15.5	111		Caution! Expired ScreenTape device

Peak Table

Size [bp]	Calibrated Conc. [ng/μl]	Assigned Conc. [ng/μl]	Peak Molarity [nmol/l]	% Integrated Area	Peak Comment	Observations
15	6.88	-	705	-		Lower Marker
173	0.237	-	2.11	1.53		
371	15.3	-	63.3	98.47		
10000	3.25	3.25	0.500	-		Upper Marker

G1: 112



Sample Table

Well	Conc. [ng/ul]	Sample Description	Alert	Observations
G1	12.3	112	⚠	Caution! Expired ScreenTape device

Peak Table

Size [bp]	Calibrated Conc. [ng/ul]	Assigned Conc. [ng/ul]	Peak Molarity [nmol/l]	% Integrated Area	Peak Comment	Observations
15	6.81	-	698	-		Lower Marker
174	0.343	-	3.03	2.79		
374	12.0	-	49.1	97.21		
10000	3.25	3.25	0.500	-		Upper Marker

Tapestation peaks and concentrations look great for all samples! All of the samples have a slight shoulder to the right, which might indicate PCR bubbles (read more [here](#)). Libraries with a bubble product can still be sequenced, which is good. I discussed with Hollie and we will move forward with sequencing. I will provide Genohub with the tapestation results, as well as the Zymo manual.